

WET LAB REAGENT LOGBOOK

Reagent Preparation, Mass Balance & Quality Control Tracking



REAG-LOG-2026-B3

Audit Log Ref: AL-86392-X

TOTAL BATCHES LOGGED

142

Quarter Q3 2026

QC APPROVED & ACTIVE

128

Ready for Assay Use

PENDING QC VERIFY

9

Hold for Sterility

EXPIRED / QUARANTINED

5

Flagged Disposal

SECTION 1: MASTER REAGENT PREPARATION RECORD – BATCH # REAG-2026-0891

Reagent Name:

10X Tris-EDTA (TE) Buffer pH 8.0

Batch / Lot ID:

REAG-2026-0891-TE80

Target Volume:

1000.0 mL (1.0 L)

Prepared By:

Dr. Eleanor Vance (Analyst)

SOP Reference:

SOP-BUFF-012 Rev 4.0

Preparation Date:

2026-07-24

Time: 09:30

Expiration Date:

2027-01-24 (6 Months)

Storage Condition:

Room Temp (20°C to 25°C)

Verified By:

Marcus Chen (QA Lead)

Quality Grade:

Molecular Biology Grade

Formulation & Chemical Mass Calculation Block

Chemical Component	CAS #	MW	Target Conc.	Calc. Mass	Weighed Mass	Vendor & Lot #
Tris Base	77-86-1	121.14	100 mM	12.114 g	12.115 g	Synapse #SY-88291
[Tris(hydroxymethyl)aminomethane]						
EDTA Disodium Salt Dihydrate	6381-92-6	372.24	10 mM	3.722 g	3.723 g	Nexa #NX-44012
Hydrochloric Acid (HCl, 6M Stock)	7647-01-0	N/A	pH Adj.	~ 4.2 mL	4.18 mL	Apex #AP-90112
Ultrapure Water (Milli-Q 18.2 MΩ · cm)	7732-18-5	18.02	Diluent	Q.S. 1000 mL	982.5 g	In-House #TANK-02

Step-by-Step Preparation Protocol Execution

Step	Procedure Specification	Check	Observed Parameter / Notes
01	Tare analytical balance (Bal-04). Weigh Tris Base (12.114 g) and EDTA (3.722 g).	[✓]	Weighed 12.115 g Tris, 3.723 g EDTA.
02	Transfer solutes into 1000 mL glass beaker. Add 800 mL ultrapure ddH ₂ O.	[✓]	Dissolved under magnetic stirring at 400 RPM.
03	Calibrate pH meter (pH-02) using pH 4.01, 7.00, and 10.01 standard buffers.	[✓]	Slope = 99.2%, Temp compensation ON (22.4°C).
04	Slowly titrate with 6M HCl under stirring until target pH 8.00 ± 0.05 is achieved.	[✓]	Final titration volume: 4.18 mL 6M HCl. pH = 8.01.
05	Transfer solution to 1000 mL volumetric flask. Q.S. to mark with ddH ₂ O.	[✓]	Meniscus aligned precisely at 1000.0 mL mark.
06	Filter sterilize through 0.22 µm PES membrane filter into sterile storage bottle.	[✓]	Autoclave indicator strip turned black. Cycle #402.

Quality Control, Verification & Sign-Off

Final Measured pH:

8.01 at 22.5°C (Spec: 8.00 ± 0.05)

Physical Appearance:

Clear, colorless, particle-free

DNase/RNase Test:

Negative (Fluorescence Assay Pass)

Analyst Signature:

Dr. Eleanor Vance

Conductivity:

14.2 mS/cm at 23.0°C

Sterility Check:

Passed (0.22 µm Filtered + Autoclaved)

Refractometer Index:

1.3342 ± 0.0005

QA Approval:

Marcus Chen, Lead QA

REAGENT BATCH IDENTIFIER BARCODE LOG



REAG-2026-0891-TE80

Scan barcode for automated LIMS inventory entry and lot genealogy tracking.

Vertex Research Campus, Bldg B-300 | CONFIDENTIAL – PROPRIETARY – GLP COMPLIANT

Page 1 of 3